NUMERICAL CONCEPTS PROBLEMS:

BY: JK BEGUMISA.

"Revise, Reflect, Achieve"

Author of; ***"Advanced Level Pure Mathematics Item Bank"*** (New curriculum) – 2025

NOTES:

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• INDICES

- **Simplify the following to the simplest possible form:**

1. $\frac{x^2(1+x^2)^{-\frac{1}{2}}-(x^2+1)^{\frac{1}{2}}}{x^2}$; $-\frac{1}{x^2\sqrt{(1+x^2)}}$

2. $\frac{(1+x)^{\frac{1}{3}}-\frac{1}{3}x(1+x)^{-\frac{2}{3}}}{(1+x)^{\frac{2}{3}}}$; $\frac{3+2x}{3(1+x)^{\frac{4}{3}}}$

3. $\frac{x^{\frac{3}{2}}+xy}{xy-y^3}-\frac{\sqrt{x}}{\sqrt{x}-y}$; $\frac{\sqrt{x}}{y}$

4. $\frac{\sqrt{(1-x)}.\frac{1}{2}(1+x)^{-\frac{1}{2}}+\frac{1}{2}(1-x)^{-\frac{1}{2}}\sqrt{(1+x)}}{(1-x)}$; $\frac{1}{(1-x)\sqrt{1-x^2}}$

5. $\frac{[2^{2n}-3.(2^{2n-3})][3^n-2.(3^{n-2})]}{3^{(n-4)}[4^{n+2}-2^{2n}]}$; $\frac{21}{8}$

6. $\frac{14^{(n+4)}+7^{(n+3)}\times 2^{(n+3)}}{13\times 14^n+14^n}$; $\mathbf{2940}$

7. Solve for x in the equation; $\left\{\left(\sqrt{\frac{16}{25}}\right)^{4x-9}\right\}^3=\frac{64}{125}$; $x=\frac{5}{2}$

8. Solve for y in the equation; $\left(\frac{1}{3}\right)^{\frac{y^2-2y}{16-2y^3}}=\sqrt[(4y)]{(9)}.$; $y=-2$

9. If; $\boldsymbol{a = 2, b = 3,}$ show that; $\boldsymbol{a^6b^4c^6\sqrt{a^{-3}b^{-4}c^6} = \left(144\sqrt{2}\right)c^9}$

10. Given that; $\boldsymbol{2^x = 4^y = 8^t}$ and $\boldsymbol{\frac{1}{2x} + \frac{1}{4y} + \frac{1}{8t} = \frac{22}{7},}$

    Clearly, show that; $\boldsymbol{x = \frac{7}{16}, y = \frac{7}{32}}$ and $\boldsymbol{t = \frac{7}{48}.}$

11. Given that; $\boldsymbol{m = 3^{\frac{2}{3}} + 3^{-\frac{2}{3}},}$ show that; $\boldsymbol{9m^3 - 27m = 82.}$

12. Given that; $\boldsymbol{x^{x\sqrt{x}} = \left(x\sqrt{x}\right)^x,}$ where $\boldsymbol{x}$ is a positive rational number, $\boldsymbol{x > 0}$.

    Find the exact value of $\boldsymbol{x.}$; $\boldsymbol{x = \frac{9}{4}.}$

13. Given that; $\boldsymbol{10^x = 3}$ and $\boldsymbol{10^y = 30.}$

    Clearly, show that; $\boldsymbol{2^x . 5^y = 15.}$

14. Solve for the **exact value** of unknown in each of the following equations;
    a) $\boldsymbol{4^{(x+4)} - 2^{(x+5)} + 1 = 0}$; $\boldsymbol{x = -4}$
    b) $\boldsymbol{2(9^x) + 3^{(-1+x)} + 4 = 2(3^{1+x})}$; $\boldsymbol{x = log_3^{(1.5)}, x = log_3^{\left(\frac{4}{3}\right)}}$
    c) $\boldsymbol{4^x \times 5^{4x+3} = 10^{2x+3}}$; $\boldsymbol{x = log_{25}^{8}}$
    d) $\boldsymbol{2^{2x+1} - 17(2^x) + 8 = 0}$; $\boldsymbol{x = -1, x = 3}$
    e) $\boldsymbol{8(4^x) - 9(2^x) + 1 = 0}$; $\boldsymbol{x = -3, x = 0}$
    f) $\boldsymbol{4^x - 2^{x+2} = 5}$; $\boldsymbol{x = log_2^5}$

15. Solve the following simultaneous equations:
    Note: *In each case, obtain the **exact values** of the unknowns*

    a) $\boldsymbol{4^x . 2^y = 128 , 3^{2x+2y} = 9^{xy}.}$
    Ans: $\boldsymbol{x = \frac{4+\sqrt{2}}{2}}$ **when** $\boldsymbol{y = 3 - \sqrt{2} , x = \frac{4-\sqrt{2}}{2}}$ **when** $\boldsymbol{y = 3 + \sqrt{2}}$

    b) $\boldsymbol{2^{(x+y+z)} = 8^{x+z-y} , 5^{3y+2} = 25^{x+z} , 3^{(2x+y+2z)} = 9^{3x+y}}$
    Ans: $\boldsymbol{x = 1, y = 2, z = 3.}$

    c) $\boldsymbol{m^{x-3} . m^{2+y} = m^2 . m^x, m^x . m^y = m}$; where; $\boldsymbol{m \neq 0, 1}$
    Ans: $\boldsymbol{x = -2, y = 3}$

    d) $\boldsymbol{\left(\sqrt{n}\right)^{x+y} = \left(\sqrt[3]{n}\right)^{y+z-1}, \left(\sqrt{t}\right)^{z-2+x} = \left(\sqrt[5]{t}\right)^{y+z}, \left(\sqrt[4]{m}\right)^{y} = \left(\sqrt[7]{m}\right)^{x+y+z};}$
    where; $\boldsymbol{n, m, t \neq 0, 1}$
    Ans: $\boldsymbol{x = 2, y = 24, z = 16}$

    e) $\boldsymbol{2^{p+q+r} = 8^{p+r-q}, 5^{3q+2} = 25^{p+r}, 3^{2p+q+2r} = 9^{3p+q}}$
    Ans: $\boldsymbol{p = 1, q = 2, r = 3.}$

16. Prove that; $\frac{16(32)^m - 2^{3m-2}.4^{1+m}}{15.(2^{m-1})(16)^m} - \frac{5^m}{\sqrt{5^{2m}}} = \mathbf{1.}$

17. Show that; $\frac{\left(9^{y+\frac{1}{4}}\right).\sqrt{3\times 3^y}}{3\sqrt{3^{-y}}} = 27^y.$

18. Clearly, prove that; $\frac{(81)^r.3^5 - 3^{4r-1}.(243)}{9^{2r}.3^3} - \frac{4(3^r)}{3^{1+r} - 3^r} = 4.$

19. If; $m = \sqrt[3]{p+q} + \sqrt[3]{p-q}$ and $p^2 - q^2 = t$, show that; $m^3 - 3mt^{\frac{1}{3}} - 2p = 0.$

20. Solve for y in the equation; $5^{3y} - 19(5^y) - 30 = 0$; $y = 1$

21. Solve for the values of x for which; $3^x + 6(3^{-x}) = 5$; $x = 1, x = log_3^2$

22. Solve the simultaneous equations: $8^{x-y} = 4^{x+y}$ and $5^{x^2-y^2} = 15625$.

    Ans: $\pm\left(\frac{5}{2}, \frac{1}{2}\right)$

23. Solve the equation; $9x^{\frac{2}{3}} + 5x^{-\frac{2}{3}} = 37$; $x = 8, x = \frac{1}{27}$

24. Solve for x in the equation: $3^{2x+1} - 26(3^x) = 9$; $x = 2$

25. Solve the equations simultaneously: $2^x + 4^y = 12$ and $3(2^x) - 2(2^y) = 16$

    Ans: $x = 3, y = 1$

26. Solve the equation; $3(3^{2x}) + 2(3^x) = 1$; $x = -1.$

27. Find the exact value of; y in the equation; $(4^{2y+1})(5^{y-2}) = 6^{1-y}$; $y = log_{480}^{37.5}$

28. Solve the equation; $9^{1+m} - 3^{3+m} - 3^m + 3 = 0$; $m = -2, m = 1$

29. Solve for; x in the equation: $2(2^x) + 3(2^{-x}) = 5$; $x = 0, x = log_2^{1.5}$

30. Show that; $\frac{\sqrt{5^{2009}}}{\sqrt{5^{2011}} - \sqrt{5^{2007}}} = \frac{5}{24}$

- **SURDS:**

31. Given that; $x = \frac{2\sqrt{24}}{\sqrt{2}+\sqrt{3}}$, prove that;

    a) $\frac{x+\sqrt{8}}{x-\sqrt{8}} = 11 + 4\sqrt{6}$

    b) $\frac{x+\sqrt{12}}{x-\sqrt{12}} = -9 - 4\sqrt{6}$

    c) $\frac{x+\sqrt{8}}{x-\sqrt{8}} + \frac{x+\sqrt{12}}{x-\sqrt{12}} = 2$

32. If; $t = \sqrt{3} + \frac{1}{\sqrt{3}}$, prove that; $t - \left(\frac{1}{t - \frac{2\sqrt{3}}{3}}\right) = \frac{5\sqrt{3}}{6}.$

33. If; $x = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ and $y = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$, show that; $\mathbf{3x^2 - 5xy + 3y^2 = 289.}$

34. If; $\mathbf{2\sqrt{54} + 2\sqrt{294} + \frac{19}{30}\sqrt{6} - \sqrt{\frac{27}{50}} - \sqrt{\frac{2}{3}} = m\sqrt{6},}$ find the value of; $\mathbf{m.}$; $\mathbf{m = 20}$

35. Given that; x and y are rational numbers such that; $\mathbf{3 + \sqrt{2} = (x + y\sqrt{2})(6 - 2\sqrt{2}),}$

    find the value of; x and y. **Ans:** $\mathbf{x = \frac{11}{14}, y = \frac{3}{7}}$

36. If; $\mathbf{2x = \left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)}$, show that; $\mathbf{\frac{\sqrt{x^2-1}}{x-\sqrt{x^2-1}} = \frac{1}{2}(t-1).}$

37. Solve the quadratic equation; $\mathbf{(-1+\sqrt{3})y^2 - (2\sqrt{3})y = (3+3\sqrt{3})}$

    Ans: $\mathbf{y = -\sqrt{3}}$ and $\mathbf{y = 3 + 2\sqrt{3}}$

38. By rationalizing the denominator of each of the following, show that;

    a) $\mathbf{\frac{4}{(1+\sqrt{2}+\sqrt{3})} = 2 + \sqrt{2} - \sqrt{6}}$

    b) $\mathbf{\frac{2}{\sqrt{5}+\sqrt{3}+2} = \frac{-1}{11}(4 + \sqrt{5} + 3\sqrt{3} - 2\sqrt{15})}$

    c) $\mathbf{\frac{1}{\sqrt{2}+\sqrt{3}-\sqrt{5}} = \frac{2\sqrt{3}+3\sqrt{2}+\sqrt{30}}{12}}$

39. Show that; $\mathbf{\sqrt{\frac{7+4\sqrt{3}}{3}} = \pm\left(1 + \frac{2}{3}\sqrt{3}\right)}$

40. Prove that; $\mathbf{\sqrt{(4+2\sqrt{3})} = \pm(1+\sqrt{3})}$

41. Solve the following simultaneous equations;

    $\mathbf{\sqrt{2}(a-1) + b\sqrt{6} = 2(1+\sqrt{3})}$ and $\mathbf{a\sqrt{6} - b\sqrt{3} = 2\sqrt{3}}$

    Ans: $\mathbf{a = 1 + \sqrt{2}}$ and $\mathbf{b = \sqrt{2}}$

42. Show, clearly that; $\mathbf{(12+6\sqrt{3})^{\frac{1}{2}} - (3-2\sqrt{2})^{\frac{1}{2}} = 4 + \sqrt{3} - \sqrt{2}}$

43. The positive constants; x and y satisfy the equation; $\mathbf{\frac{\sqrt{x}}{2x+\sqrt{x}} = \frac{2\sqrt{x}-y}{3x+y}.}$

    Show that; $\mathbf{y = \frac{x+2\sqrt{x}}{2(1+\sqrt{x})}.}$

44. Solve the following pairs of simultaneous equations;

    a) $\mathbf{x\sqrt{2} + y\sqrt{3} = 5}$, $\mathbf{(5\sqrt{3}-\sqrt{2})x + (5\sqrt{2}-\sqrt{3})y = 10\sqrt{6}}$

       Ans: $\mathbf{x = \sqrt{2} + \sqrt{3}}$, $\mathbf{y = \sqrt{3} - \sqrt{2}}$

    b) $\mathbf{(x + y\sqrt{3})^2 = (56 + 12\sqrt{3})}$, $\mathbf{y = 3x}$

       Ans: $\mathbf{x = \sqrt{2}}$ when $\mathbf{y = 3\sqrt{2}}$, $\mathbf{x = -\sqrt{2}}$ when $\mathbf{y = -3\sqrt{2}}$

45. Clearly, show that; $\mathbf{(1+\sqrt{3})^4 = 28 + 16\sqrt{3}}$

46. Solve for $\boldsymbol{x}$ in the equation; $\sqrt{3}(-\sqrt{3}+x) = x + \sqrt{3}$.

    Ans: $x = 3 + 2\sqrt{3}$

47. Prove that; $\frac{(2+\sqrt{3})^2-(1-\sqrt{3})^2}{\sqrt{3}} = 6 + \sqrt{3}$

48. Solve the equation; $8w^{\frac{1}{2}} - w^{-1} = 0$.

    Ans: $w = \frac{1}{4}$

49. Given that; $p = \frac{3}{2}$, $q = \frac{9-\sqrt{17}}{4}$ and $r = \frac{9+\sqrt{17}}{4}$.

    Prove that; $p + q + r = pqr$.

• LOGARITHMS:

- **Prove out the following logarithmic laws:**

50. $log_a^x + log_a^y = log_a^{xy}$

51. $log_b^m - log_b^n = log_b^{\left(\frac{m}{n}\right)}$

52. $log_m^p = \frac{log_y^p}{log_y^m}$

53. $log_r^p = \frac{1}{log_p^r}$

54. $b^{(log_b^w)} = w$

55. $y^{(log_c^p)} = p^{(log_c^y)}$

56. $log_{y^n}^x = \frac{1}{n} log_y^x$

- **Solve for unknown in each of the following logarithmic equations.**

57. $1 + 2log_5^t = log_5^{(-3+16t)}$; $t = 3, t = \frac{1}{5}$.

58. $log_y^{27} = 3 + log_y^8$; $y = \frac{3}{2}$

59. $log_2^y + log_2^{(3y+4)} = 2log_2^{(3y-4)}$; $y = 4$

60. $2log_3^h - log_3^{(h-2)} = 2$; $h = 3, h = 6$

61. $log_2^{(7w-1)} = 3 + log_2^{(w-1)}$; $w = 7$

62. $log_2^{(y-1)} - log_2^{(y-3)} = log_2^{\left(\frac{1}{y}\right)}$; ***No real value of y***

63. $2log_a^x = log_a^{18} + log_a^{(-4+x)}$; $x = 6, x = 12$

64. $log_x^8 - log_{x^2}^{16} = 1$; $x = 2$

65. $(log_x^4)(log_8^x) = log_{27}^x$; $x = 9$

66. $\left(log_3^{x^2}\right)(log_{9x}^3) = 1$; $x = 9$

67. $log_{10}^{(4+19x^2)} - 2log_{10}^x = 2$; $x = \frac{2}{9}$

68. $log_{(2-x)}^{(2x^2-1)} = 2$; $\boldsymbol{x = -5}$

69. $\sqrt{(log_2^x)} + 3log_2^{x^{\frac{1}{2}}} = 4$; $\boldsymbol{x = 2^{\frac{16}{9}}.}$

70. $log_{10}^{\left(\frac{y^2+24}{y}\right)} = 1$; $\boldsymbol{y = 4, y = 6}$

71. $3^{(log_9^4)} + 5^{(3log_5^2)} = 6^{(log_6^x)}$; $\boldsymbol{x = 10}$

72. $log_{25}^{4x^2} = log_5^{(3-2x^2)}$; $\boldsymbol{x = 0.823}$

73. $log_4^{(5-4x)} = log_2^x$; $\boldsymbol{x = 1}$

74. $log_x^{27} - log_{x^2}^{81} = 1$; $\boldsymbol{x = 3}$

75. $3log_7^x = 2log_x^7 + 5$; $\boldsymbol{x = 7^{-\frac{1}{3}}, x = 49}$

76. $log_x^5 + 5log_5^x = 6$; $\boldsymbol{x = 5^{\frac{1}{5}}, x = 5}$

77. $3log_2^x - log_x^2 = 2$; $\boldsymbol{x = 2, x = 2^{-\frac{1}{3}}}$

78. $log_2^x - log_x^8 = 2$; $\boldsymbol{x = \frac{1}{2}, x = 8}$

79. $2log_5^{(3^m-4)} = log_5^{16} + log_5^{\left(3^m+\frac{97}{16}\right)}$; $\boldsymbol{m = 3}$

80. $log_4^{p^3} + log_2^{\sqrt{p}} = 8$; $\boldsymbol{p = 16}$

81. $log_{16}^{r^5} = log_{64}^{125} - log_4^{\sqrt{r}}$; $\boldsymbol{r = 5^{\frac{1}{3}}}$

82. $5 \times 5^{logx} + 5^{2-logx} = 30;\quad x > 0$; $\boldsymbol{x = 1, x = 10}$

83. $\frac{log_4^{x^2}}{5+log_4^{x^2}} + (log_4^x)^2 = 0$; $\boldsymbol{x = 1, x = \frac{1}{2}, x = \frac{1}{16}}$

84. $\frac{2-log_4^{x^7}}{7-log_4^{x^2}} + (log_4^x)^2 = 0$; $\boldsymbol{x = 2, x = 4, x = 16}$

85. $\frac{x^{(log_2^x)}}{x^2} = 8$; $\boldsymbol{x = \frac{1}{2}\ ,\ x = 8}$

86. $3^{(logx)} + 3 \times x^{log3} = 36$; $\boldsymbol{x = 100}$

87. $ln(e^{2x} + 8) = 2x + ln3$; $\boldsymbol{x = ln2}$

88. $log_2^{(-1+x)} - log_2^{(3+x)} = log_2^{\left(\frac{1}{x}\right)}$; $\boldsymbol{x = 3}$

89. $log_4^{(48+2^t)} + 1 = t$; $\boldsymbol{t = 4}$

90. $log_3^{(1+x)} = log_{(x+1)}^3$; $\boldsymbol{x = -\frac{2}{3}, x = 2}$

91. $(3x)^{log3} = (4x)^{log4}$; $\boldsymbol{x = \frac{1}{12}}$

92. $log_2^{(11-6x)} = 3 + 2log_2^{(-1+x)}$; $\boldsymbol{x = \frac{3}{2}}$

93. $2log_2^x + 5log_x^2 = 7$; $\boldsymbol{x = 2 , x = 4\sqrt{2}}$

94. $log_3^{(2x+1)} - log_3^{(3x+4)} + log_3^{(3x-2)} = log_3^x$; $\boldsymbol{x = 2}$

95. $2log_2^{\left(\frac{y}{2}\right)} + log_2^{(\sqrt{y})} = 8.$; $\boldsymbol{y = 16}$

96. $log_2^{(3x-4)} = \frac{1}{3}log_2^{(8x^6)} - log_2^4$; $\boldsymbol{x = 4,} x = 2$

97. $log_3^2 + 2log_3^x = \frac{1}{log_{(7x-3)}^3}$; $\boldsymbol{x = \frac{1}{2}, \ x = 3.}$

98. $log\left\{\frac{1+m}{1-m}\right\} = log\left\{\frac{m+3}{4-m}\right\}$; $\boldsymbol{m = -\frac{1}{5}.}$

99. $\log(2+x) + \log(-1+x) = 1$; $\boldsymbol{x = 3}$

100. $log_5^{\left(\frac{y}{\sqrt{y}}\right)} = \frac{2}{log_5^y}$; $\boldsymbol{y = 25, y = \frac{1}{25}}$

101. $3log_8^x = 2log_x^8 + 5$; $\boldsymbol{x = \frac{1}{2}, x = 64}$

102. $log_4^x = 1 - log_4^{(-3+x)}$; $\boldsymbol{x = 4}$

103. $log_2^x + log_x^8 = 4$; $\boldsymbol{x = 2 , x = 8}$

104. $16 = x^{log_2^x}$; $\boldsymbol{x = 4 , x = \frac{1}{4}}$

105. $log_2^5 + log_2^t = log_2^{(20+t)}$; $\boldsymbol{t = 5}$

106. $-5 + log_2^m + log_m^{64} = 0$; $\boldsymbol{m = 4 , m = 8}$

107. $log_9^{(-5+21x)} = log_3^{(2x)}$; $\boldsymbol{x = \frac{1}{4}, x = 5}$

108. $log_x^8 - log_{x^2}^{16} = 1$; $\boldsymbol{x = 2}$

109. $3^{log_3^2} + 5^{log_5^7} = 6^{2log_6^m}$, where; $m > 0$; $\boldsymbol{m = 3}$

110. $11^{log_d^7} = 7$; $\boldsymbol{d = 11}$

111. $log_2^y + log_4^y + log_{16}^y = \frac{21}{4}$; $\boldsymbol{y = 8}$

112. $ln4 + 2lnx - \ln(x + x^2) = ln2$, $x > 0$; $\boldsymbol{x = 1}$

113. $\ln(x^2 - 1) - \ln(x - 1) = ln5$, where; $x > 0$; $\boldsymbol{x = 4}$

114. $log_a^x + log_a^{(x-3)} = log_a^{10}$; $\boldsymbol{x = 5}$

115. $log_2^y = 9log_y^2$; $\boldsymbol{y = 8, y = \frac{1}{8}}$

116. $log_2^{(6-x)} = 3 - log_2^x$; $\boldsymbol{x = 4, x = 2}$

117. $log_a^{(1+\sqrt{y})} = \frac{1}{2}log_a^{(9+\sqrt{16y})}$; $\boldsymbol{y = 16}$

118. $2log_2^x + log_2^{(x-1)} - log_2^{(5x+4)} = 1$; $\boldsymbol{x = 4.}$

119. $log_4^{(6-x)} = log_2^x$; $\boldsymbol{x = 2}$

120. $log_m^{(x+3)} + \frac{1}{log_x^m} = 2log_m^2$; $\boldsymbol{x = 1}$

121. $x - \{xlog5\} = log6 - log(1 + 2^x)$; $\boldsymbol{x = 1.}$

122. $log_{25}^x + log_{125}^x + log_{625}^x = \frac{13}{2}$; $\boldsymbol{x = 5^6}$

123. $3 + log_3^d = 4log_d^3$; $\boldsymbol{d = \frac{1}{81}, d = 3}$

124. $1 + 2log_4^t = log_t^4$; $\boldsymbol{t = 2}$

125. $log_{49}^{7w} + log_7^{49w} = 7$; $\boldsymbol{w = 7^3}$

- **Solve each of the following sets of simultaneous logarithmic equations:**

Note: *In each case, obtain the* ***exact*** *values of the unknowns.*

126. $log_2^{(p^2m)} = 2$, $11 + \frac{1}{2}log_2^m = 3log_2^p$; $\boldsymbol{p = 8, m = \frac{1}{16}}$

127. $2log_2^x - log_2^y = 1$, $log_2^{(4x\sqrt{y})} = 1$; $\boldsymbol{x = 2^{-\frac{1}{4}}, y = 2^{-\frac{3}{2}}}$

128. $log_4^x - log_9^y = 0$, $log_2^x + log_3^y = 3$; $\boldsymbol{x = 2\sqrt{2}\,, y = 3\sqrt{3}}$

129. $y^{logx} = 100$, $log\left\{\sqrt{\frac{xy}{10}}\right\}$; $\boldsymbol{x = 10, y = 100}$ ***, or the other way round.***

130. $7^y.\left(\frac{1}{49}\right)^x = 1$, $log_{\sqrt{2}}^{(2x+3y)} - log_{\sqrt{2}}^2 = 4$; $\boldsymbol{x = 1, y = 2}$

131. $\log(2 - x) - logy = 1 - log2$, $log2x = logy$; $\boldsymbol{x = \frac{2}{11}, y = \frac{4}{11}}$

132. $log_3^x + log_3^y = 2$, $log_3^{(2x-3)} - log_3^y = 1$; $\boldsymbol{x = \frac{9}{2}, y = 2}$

133. $\log(f + g) = 0$, $2logf = \log(1 + g)$; $\boldsymbol{f = 1\,, g = 0}$

134. $2log_n^m + 2log_m^n = 5$, $mn = 8$; $\boldsymbol{n = 2, m = 4,}$ ***or the other way round.***

135. $log_2^{xy} = 7$, $log_2^{\left(\frac{x^2}{y}\right)} = 5$; $\boldsymbol{x = 16, y = 8}$

136. $6log_3^x - 2log_3^y = 7$, $2log_3^x - 4log_3^y = 9$; $\boldsymbol{x = \sqrt{3}\,, y = \frac{1}{9}}$

137. $log_3^{pq} = 4$, $log_3^{\left(\frac{p}{q}\right)} = 2$; $\boldsymbol{p = 27}$ **when** $\boldsymbol{q = 3, p = -27}$ **when** $\boldsymbol{q = -3.}$

138. $log_p^{(t+s)} = 0$, $2log_p^t = log_p^{(1+4s)}$; $\boldsymbol{t = 1\,, s = 0}$

139. $log_2^{(x-14y)} = 3$, $log_{10}^x - log_{10}^{(1+y)} = 1$; $\boldsymbol{x = 15, y = \frac{1}{2}}$

140. $log_2^w + 2log_4^y = 4$, $7w - y = 6$; $\boldsymbol{w = 2,\ y = 8}$
141. $6log_3^x + 6log_{27}^y = 7$, $4log_9^x + 4log_3^y = 9$; $\boldsymbol{x = \sqrt{3}}$ **and** $\boldsymbol{y = 9}$
142. $2log_x^y = 1$, $xy = 64$; $\boldsymbol{x = 16 , y = 4}$
143. $ylog_2^8 = x$, $2^x + 8^y = 8192$; $\boldsymbol{x = 12 , y = 4}$
144. $logx - log2 = 2logy , x - 5y + 2 = 0$; $\boldsymbol{\left(\frac{1}{2} , \frac{1}{4}\right)}$ **and** $\boldsymbol{(8 , 2)}$
145. $log_2^x + 2log_4^y = 4 , 12y + x = 52$; $\boldsymbol{(4 , 4)}$ **and** $\boldsymbol{\left(48 , \frac{1}{3}\right)}$
146. $ln6 + ln(x - 3) = 2lny$, $2y - x = 3$; $\boldsymbol{x = 9 , y = 6}$
147. $2 + log_2^{(1+2x)} = 2log_2^y$, $x = 22 - y$; $\boldsymbol{x = 12 , y = 10}$
148. $8^y = 4^{2x+3}$, $log_2^y = 4 + log_2^x$; $\boldsymbol{x = \frac{3}{22}, y = \frac{24}{11}}$
149. $(25)^x.(5)^y = 0.2$, $log_2^{(x-y)} = 2 + log_2^{(4+y)}$; $\boldsymbol{x = 1, y = -3}$
150. $2log_3^x - 3log_3^y = 7$, $log_3^x - 2log_3^y = 4$; $\boldsymbol{x = 9, y = \frac{1}{3}}$
151. $2log_x^y = 1$, $xy = 64$; $\boldsymbol{x = 16, y = 4}$
152. $log_x^y = 2$, $2logy = log2 + 3logx$; $\boldsymbol{x = 2, y = 4}$
153. $logx + 2logy = 1$, $x + 3y^2 = 13$; $\boldsymbol{x = 10}$ **when** $\boldsymbol{y = 1, x = 3}$ **when** $\boldsymbol{y = \frac{1}{3}\sqrt{30}}$
154. $log_2^{(-1+y)} = 1 + log_2^x$, $2log_3^y = 2 + log_3^x$; $\boldsymbol{\left(\frac{1}{4} , \frac{3}{2}\right)}$ **and** $\boldsymbol{(1 , 3)}$
155. $3log_8^{xy} = 4log_2^x$, $log_2^y = 2 + log_2^x$; $\boldsymbol{x = 2}$ **and** $\boldsymbol{y = 8}$
156. $x^{logy} = 2x - 5$, $y^{logx} = x + 5$; $\boldsymbol{x = 10, y = 15}$
157. $logx + 10^{logy} = 7$, $x^y = 10^{12}$; $\boldsymbol{(10^4 , 3)}$ **and** $\boldsymbol{(10^3 , 4)}$
158. $log_2^{(xy^2)} = 0 , log_2^{(x^2y)} = 3$; $\boldsymbol{x = 4, y = \frac{1}{2}}$.
159. $log_2^{(3x+4)} = 1 + log_2^y$, $2log_2^y = \frac{3}{log_x^2}$; $\boldsymbol{x = 4, y = 8}$
160. $log_2^x + 2log_4^y = 4, x + y = 10$; $\boldsymbol{(8, 2),}$ ***or the other way round.***
161. $log_y^x = 5$, $log_2^x = 2 + log_2^y$; $x = \boldsymbol{4\sqrt{2}, y = \sqrt{2}}$
162. $2logy = log2 + logx , 2^y = 4^x$; $\boldsymbol{x = \frac{1}{2}, y = 1}$
163. $log_2^{(x+y)} = 0 , log_2^x - \frac{2}{log_y^2} = 1$; $\boldsymbol{x = y = \frac{1}{2}}$

164. Given that; $\boldsymbol{2log_2^{(x+15)} - log_2^x = 6.}$

a) Show that; $\boldsymbol{x^2 - 34x + 225 = 0.}$

b) Hence or otherwise, solve the equation; $2log_2^{(x+15)} - log_2^x = 6.$

165. Given that; $log_2^{\left(x^{\frac{1}{4}}\right)} + log_2^{y^4} = 2 - \frac{1}{log_8^{64}}$

Show that; $y^{16} = \frac{64}{x}.$

166. Two curves; $y = 8^x$ and $f(x) = 2(3^x)$ intersect at point T.

a) Show, by solving, that at point T, $x = \frac{1}{2-log_2^3}$

b) Hence, find the **other exact coordinate value** of point T.

167. In $x - y$ plane, two curves are defined by the equations;

$$f(x) = 9^x \text{ and } g(x) = 6(5^x)$$

Show, by solving, that at their point of contact, $x = \frac{1+log_3^2}{2-log_3^5}.$

168. Given two simultaneous equations:

$$4p - \frac{1}{2}q = log_6^{(3.2)} \text{ and } q - p + 1 = log_6^{(75)}.$$

a) By clear solving, show that; $p = \frac{1}{1+log_2^3}.$

b) Hence, or otherwise, find the exact value of; q, in log raised to base two.

169. If; $m^2 + n^2 = 25mn$, then, prove that;

$$2log(m+n) = 3log3 + logm + logn.$$

170. Given that; $x^2 + y^2 = r^2$. Prove that; $log_y^{(r+x)} + log_y^{(r-x)} = 2.$

171. If; $x^4 + y^4 = 79x^2y^2$, prove that;

a) $log\left(\frac{x^2+y^2}{9}\right) = logx + logy$

b) $2log_{11}^{(x+y)} = 1 + log_{11}^x + log_{11}^y$

172. Prove that; $2log\left(\frac{35}{192}\right) + 2log\left(\frac{114}{91}\right) + log48 + 2log\left(\frac{13}{19}\right) = log\left(\frac{75}{64}\right)$

173. If; $log_3^2, log_3^{(2^x-5)}$ and $log_3^{\left(2^x-\frac{7}{2}\right)}$ are in Arithmetic Progression,

find the exact value of; $x.$; $x = 3$

174. If; $a^2 + b^2 = 7ab$, prove that; $log\left\{\frac{1}{3}(a+b)\right\} = \frac{1}{2}(loga + logb).$

175. Prove that; $log2 + 16log\left(\frac{16}{15}\right) + 12log\left(\frac{25}{24}\right) + 7log\left(\frac{81}{80}\right) = 1.$

176. If; $log(x^3y^3) = m$ and $log\left(\frac{x}{y}\right) = n,$

by solving those two equations simultaneously, show that;

a) $x = 10^{\frac{1}{6}(m+3n)}$

b) $y = 10^{\frac{1}{6}(m-3n)}$

177. If; $x^3 + y^3 = 0,$ where; $x + y \neq 0,$ prove that;

$log(x + y) = \frac{1}{2}(log3 + logx + logy)$

178. If; $a^x = b^y = c^t$ and $b^2 = ac,$ prove that; $y = \frac{2xt}{x+t}.$

179. Given that; $z = log_r^{(mn)}, x = log_m^{(rn)}$ and $y = log_n^{(rm)}.$

Prove that; $xyz = 2 + x + y + z.$

180. Given that; $x = log_a^N$ and $y = log_b^N.$

Prove that; $log_{ab}^N = \frac{xy}{x+y}$

181. If; x and y are such that; $x^{(x+y)} = y^{12}$ and $y^{(x+y)} = x^3.$

Prove that; $logx = 2logy.$ Hence, find the exact values of; x and $y.$

$x = 9$ when $y = -3$, $x = 4$ when $y = 2$

182. Given that; $a, y, b \neq 0$ and; $2 + log_a^b + 3log_a^y = 2log_a^{(a^2y)}.$

By solving, show that; $y = \frac{a^2}{b}.$

183. Given that; log_a^x , log_b^x and log_c^x are in Arithmetic Progression,

Show that; $c^2 = (ac)^{log_a^b}$

184. Given that; $m = log_7^{12}$ and $r = log_{12}^{24}.$

Express; log_{54}^{168}, in terms of; m and $r.$

185. Given that; log_5^{16} , $log_5^{(3^x-4)}$, $log_5^{\left(3^x+\frac{97}{16}\right)}$ are in AP, find the exact value of $x.$

186. Given that; $log2 = t$, prove that; $log_8^5 = \frac{1-t}{3t}.$

187. Given that; $x = log_2^3$, and $y = log_4^5$, show that; $log_{45}^2 = \frac{1}{2(x+y)}$

188. Given that; $p = log_a^4$ and $q = log_a^5,$ express each of the following logs in terms of p and q;

a) log_a^{100} ; $p + 2q$

b) $log_a^{0.4}$; $\frac{1}{2}p - q$

189. Given that; $m = log_2^3$ and $t = log_2^5,$ express each of the following logs in terms of m and $t;$

a) $\boldsymbol{log_2^{45}}$; $\boldsymbol{2m+t}$

b) $\boldsymbol{log_2^{0.3}}$; $\boldsymbol{m-t-1}$

190. Given that $\boldsymbol{k}$ satisfies the equation; $\boldsymbol{log_k^{(1+5y)} = 4 + \frac{1}{log_3^k}}$.

Solve for $\boldsymbol{y}$ in terms of $\boldsymbol{k}$. ; $\boldsymbol{y = \frac{3k^4-1}{5}}$

191. Given the equation; $\boldsymbol{(a^4 - 2a^2b^2 + b^4)^{x-1} = (a-b)^{2x}(a+b)^{-2}}$

Solve for exact value of $\boldsymbol{x}$ in terms of $\boldsymbol{a}$ and $\boldsymbol{b}$. ; $\boldsymbol{x = \frac{log(a-b)}{\log(a+b)}}$

192. Show that if; $\boldsymbol{loga, logb}$ and $\boldsymbol{logc}$ are consecutive terms of an Arithmetic Progression, then; $\boldsymbol{a, b}$ and $\boldsymbol{c}$ are also in Geometric Progression.

193. Show that; $\boldsymbol{3log(a+b) = 3logb + log\left\{1 + \frac{3a}{b}\left(1+\frac{a}{b}\right) + \frac{a^3}{b^3}\right\}}$

194. Given that; $\boldsymbol{log_5^{21} = m}$ and $\boldsymbol{log_9^{75} = n}$, show that; $\boldsymbol{log_5^7 = \frac{1}{2n-1}(2nm - m - 2)}$.

195. Given that; $\boldsymbol{log_9^{xy} = 6}$, prove that; $\boldsymbol{log_3^x + log_3^y = 12}$.

Hence, solve simultaneous equations; $\boldsymbol{log_9^{xy} = 6}$ and $\boldsymbol{(log_3^x)(log_3^y) = 20}$.

Ans: $\boldsymbol{(3^{10}, 9)}$, ***or vice versa.***

196. Prove that; $\boldsymbol{log_a^{(a^mbc)} = \frac{m}{log_{bc}^a}\left\{\frac{1}{m} + log_{bc}^a\right\}}$

197. Given that; $\boldsymbol{log_2^x + 2log_4^y = 4}$. Show that; $\boldsymbol{xy = 16}$

198. Given that; $\boldsymbol{\frac{log3}{log2} = \frac{8}{5}}$, solve for; $\boldsymbol{x}$ and $\boldsymbol{y}$ in the simultaneous equations;

$\boldsymbol{3^x = 2^{3y+1}}$ and $\boldsymbol{4^{x-1} = 12^{2y+1}}$; $\boldsymbol{x = -\frac{40}{23}, y = -\frac{29}{23}}$

199. Find the value(s) of $\boldsymbol{p}$ such that the equation; $\boldsymbol{log(py) = 2log(y+1)}$ has exactly one root. ; $\boldsymbol{p = 1, p = 3}$.

200. Solve the simultaneous equations: $\boldsymbol{log_2^x - log_4^y = 4}$, $\boldsymbol{log_2^{(x-2y)} = 5}$

Ans: $\boldsymbol{x = 64}$, $\boldsymbol{y = 16}$

If Not We, Then Who?

If Not Now, Then When?

"WISHING ALL A – LEVEL STUDENTS OF 2026 COUNTLESS SUCCESS"

END